

Empowering CG ProductionCost-Effective Techniques for Voluminous Fur Rendering with Unreal Engine

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Abstract

This paper presents a novel approach to optimizing realistic fur rendering for CG animation using Unreal Engine (UE). We introduce a progressive method combining three key techniques to enhance rendering efficiency while maintaining high quality. These strategies significantly optimize GPU memory use, enabling more complex and realistic results. Our approach streamlines production workflows, reducing both time and costs, offering broader potential applications in the film industry.

CCS Concepts

• **Computing methodologies** → **Rendering**; *Shape modeling*.

Keywords

fur rendering, GPU memory optimization, production pipeline

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1 Introduction

Rendering realistic fur in CG animation production presents significant challenges, particularly in terms of cost and time. Techniques such as real-time shading optimization in games [Patry 2020] and BRDF-based fur shading in films [Pieké et al. 2020] have improved rendering efficiency and quality under hardware constraints. Recent advancements in GPU-supported Real-Time (RT) engines and path tracing have further accelerated rendering while maintaining quality. Our tests show that GPU-based path tracing reduces frame rendering times by 88%, compared to CPU rendering (Figure 1). However, large-scale production still faces resource limitations, particularly in GPU capabilities and video memory, often constrained by production budgets.

Recognizing this challenge as an opportunity, this paper introduces a novel approach that combines three strategies—node-splitting, mesh conversion, and density increase—to optimize fur



	CPU(I7-10700)	GPU(RTX 4090)
RESOLUTION	HD 1080P	HD 1080P
SAMPLE	4/4/4/2/3/3	Spatial Sample 256;Temporal Sample 1
RENDER TIME	26mins	3mins

Figure 1: GPU-based path tracing cuts frame rendering times by 88%.

rendering efficiency without compromising quality. These methods enhance production workflows and reduce costs, making them especially beneficial for projects with extensive fur content.

2 Approach and Implement

Our approach builds progressively, addressing specific issues and enhancing overall rendering quality and efficiency.

2.1 Nodes-splitting

We first tackle memory overflow during detailed fur rendering on individual characters. Using a single NVIDIA GeForce RTX 4090 GPU at 1080p, we rendered high-quality fur with over 1.4 million strands, replicating plush textures of characters. However, this pushed the GPU's memory limits, causing artifacts (Figure 2).



Figure 2: Extra lines appeared around the tail area, due to rendering errors.

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To resolve this, we split the fur into multiple groom nodes, each holding fewer than 600,000 strands (Figure 3), and applied them to the Skeletal Mesh Component in UE, ensuring more even allocation of GPU memory and smoother rendering.

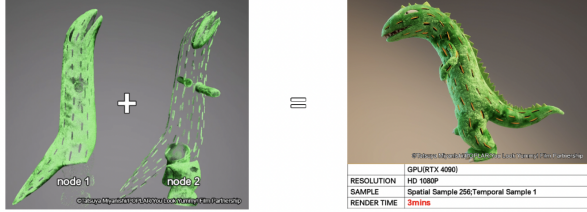


Figure 3: High-quality character rendered after splitting groom nodes 1 and 2, each containing less than 600,000 strands.

2.2 Mesh Conversion

A key bottleneck in rendering is the limit on fur strands a scene can accommodate without exceeding GPU memory. To address this, we propose mesh conversion, inspired by the neural radiance cache concept [Müller et al. 2021].

Drawing on extensive project experience, we concluded that elements like trees, bushes, grass, and props do not require editable fur; they only need to appear fur-covered. The focus should be on the fur attached to characters. Specifically, we convert non-essential elements into simplified, static meshes that mimic fur, while only "skin" assets on characters retain detailed fur rendering. This reduces computational complexity and GPU memory usage, freeing up resources for more detailed rendering on main characters, enhancing realism and visual richness. Stress testing shows that mesh conversion doubled the scene's capacity, increasing renderable fur strands from 24.8 million to 49.5 million, setting a new benchmark for GPU-based rendering.

2.3 Density Increase

Following mesh conversion, we extended static meshes across larger environments by using UE's Foliage system and the Procedural Content Generation (PCG) Framework. By applying density increase techniques, the scene's fur strand capacity dramatically increased from 49.5 million to 955.5 million (Figure 4), enabling more efficient enhancement of scene complexity.

3 Discussion and Future Work

In conclusion, the outlined strategies optimize computational resources, reduce production times, and lower costs, marking a significant advancement in CG animation pipelines. Our calculations indicate that these methods could reduce the Lighting, Rendering, and Compositing (LRC) cycle by 81%, shortening episode production by 44%, or by 7-weeks in a typical 16-week schedule.

Furthermore, UE's rapid rendering capabilities enable near-final visuals during pre-visualization, providing early insights and supporting informed decision-making throughout content development.

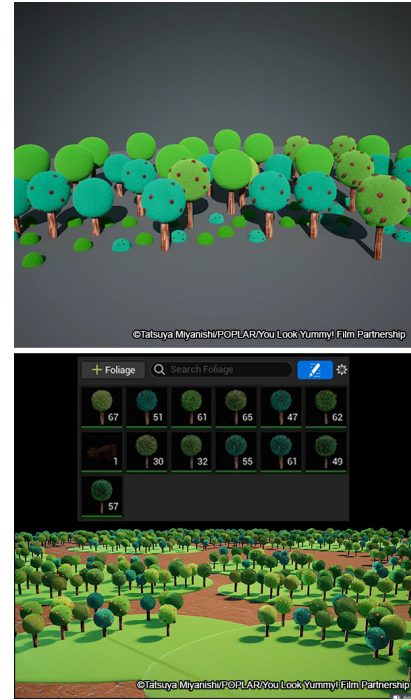


Figure 4: Top: Mesh conversion for flora in the environment, showing 49.5 million fur strands, detailing 30 trees (approximately 1.5 million strands each) and 30 shrubs (0.15 million strands each). Bottom: Visualization of 637 trees, totaling 955.5 million fur strands after applying density increase, with each tree containing approximately 1.5 million strands.

Future improvements will focus on enhancing fur realism, such as incorporating anisotropic highlights, and developing metrics to evaluate realism, complementing our workflow optimization.

We are committed to integrating these strategies into our pipelines, with plans to extend this work to the film industry.

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